

GEOGRAPHY

PHYSICAL GEOGRAPHY



CLIMATOLOGY

MODULE – 33

WORLD CLIMATIC TYPES
THE HOT DESERT AND MID-LATITUDE
DESERT CLIMATES (PART-1)

WORLD CLIMATIC TYPES

3

HOT DESERT AND MIDLATITUDE DESERT CLIMATES

DISTRIBUTION

- 1 Deserts are regions of scanty rainfall, which may be hot like **hot deserts of Sahara** or temperate as are the **mid latitude deserts like the Gobi Desert**.



- 2 The aridity of the hot deserts is mainly due to the **effects of offshore trade winds**. Hence, they are also called **as trade wind deserts**.
- 3 The temperate deserts are rainless because of their **interior locations** in the temperate latitudes well away from the rain bearing winds.
- 4 The major hot deserts of world are located on the **west coast of the continents between the latitudes of 20⁰-30⁰ North and South**.

ATMOSPHERIC PRESSURE

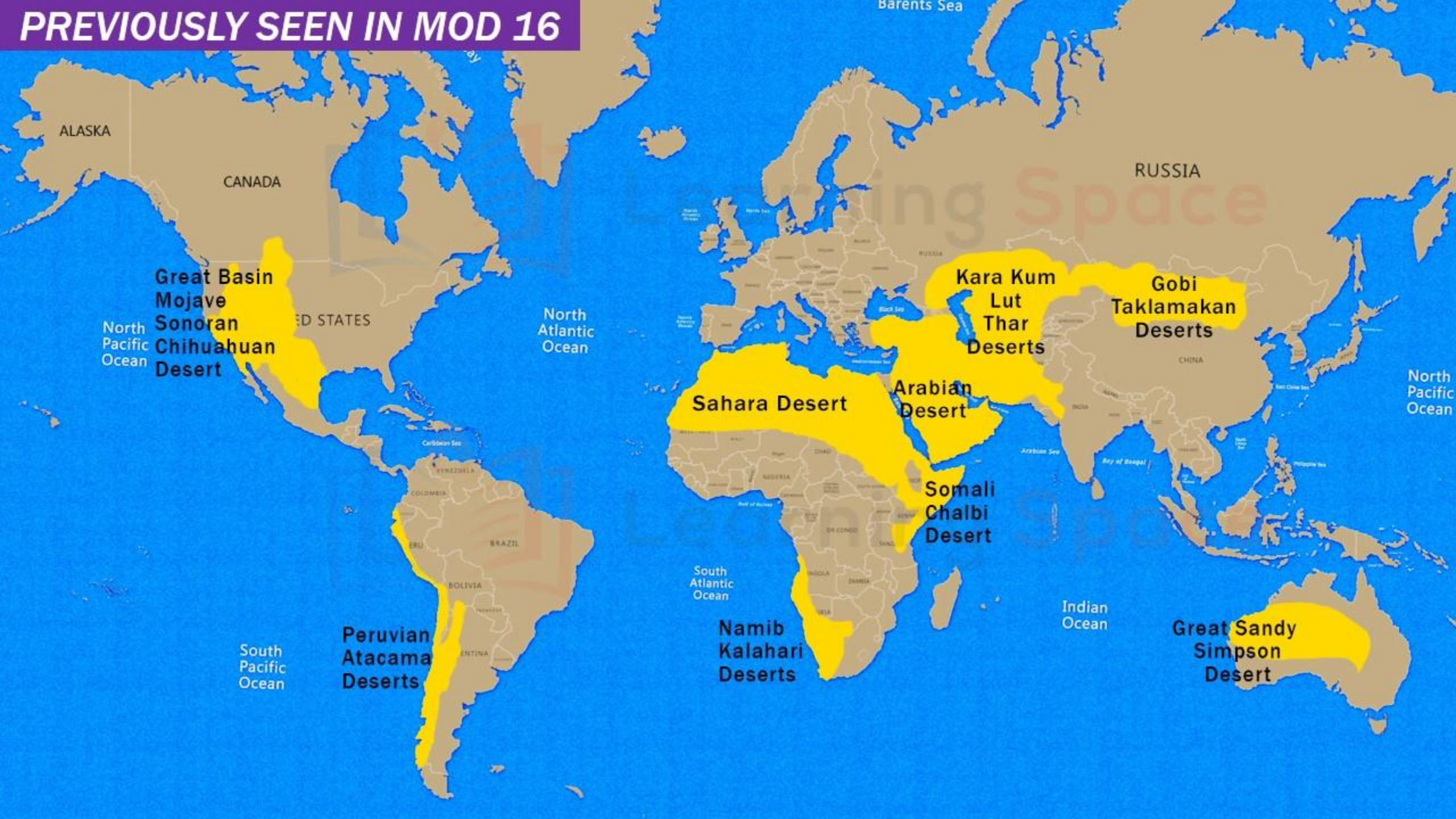
UPSC 2013 GS-1 QUESTION

Major hot deserts in northern hemisphere are located between 20-30 degree north and on the western side of the continents. Why? (10 Marks)

HINTS

- 1 Offshore trade winds in the region
- 2 Prevailing anticyclonic conditions- Areas between 20-30 degree latitudes on western margins of continents are the regions of descending air
- 3 Presence of cold ocean currents along the western coast of continents- this stabilizes the air and prevents cloud formation and rainfall.
- 4 Leeward sides of mountains/Parallel mountain ranges

PREVIOUSLY SEEN IN MOD 16

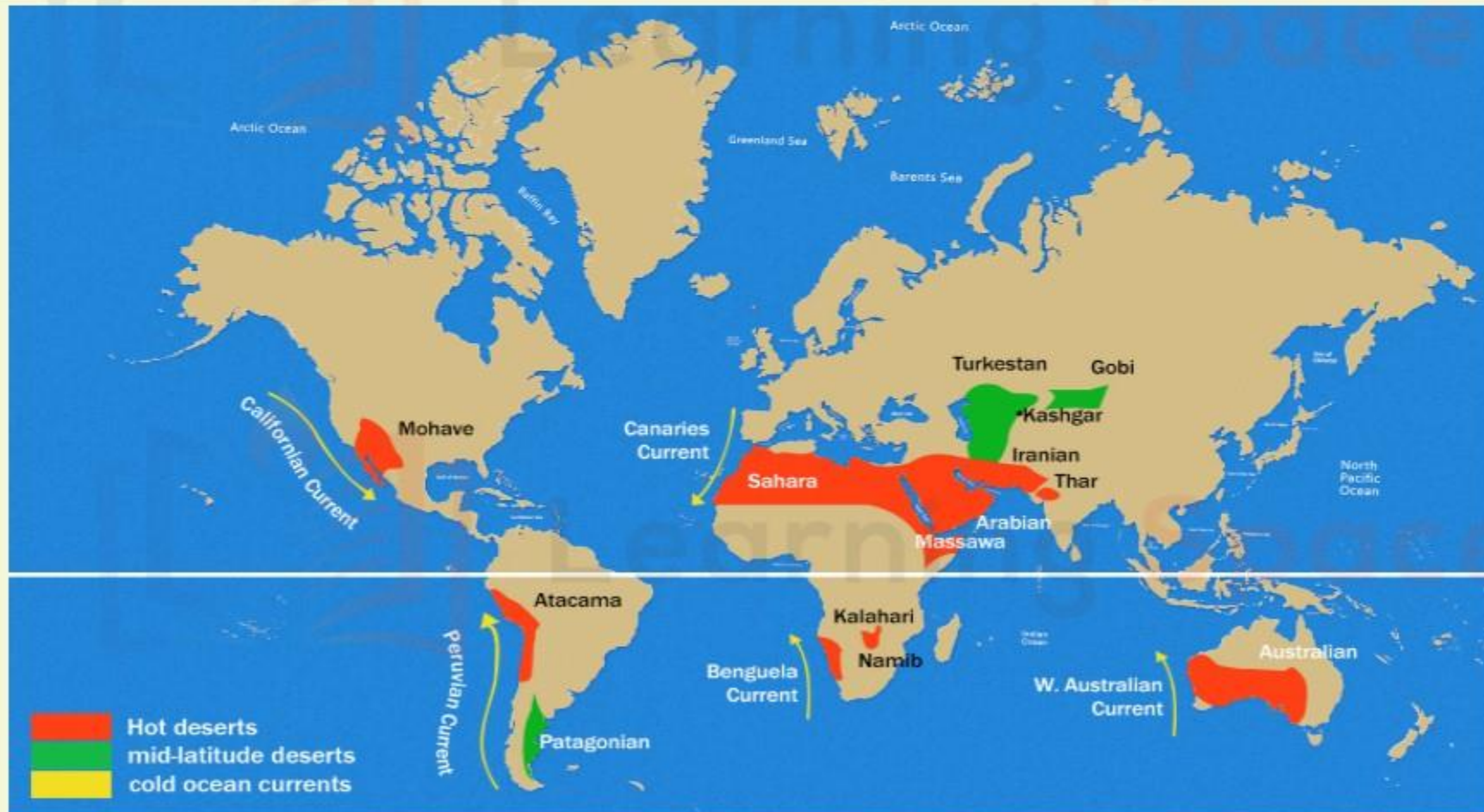


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HOT DESERT AND MIDLATITUDE DESERT CLIMATES DISTRIBUTION

4

They include the Sahara Desert, the Australian Desert, the Arabian Desert, Iranian Desert, Thar Desert, Kalhari and Namib Desert.



3

HOT DESERT AND MIDLATITUDE DESERT CLIMATES

DISTRIBUTION

6 In South America, the Atacama and Peruvian Desert is the driest of all the deserts with less than 0.5 inches of rainfall annually.



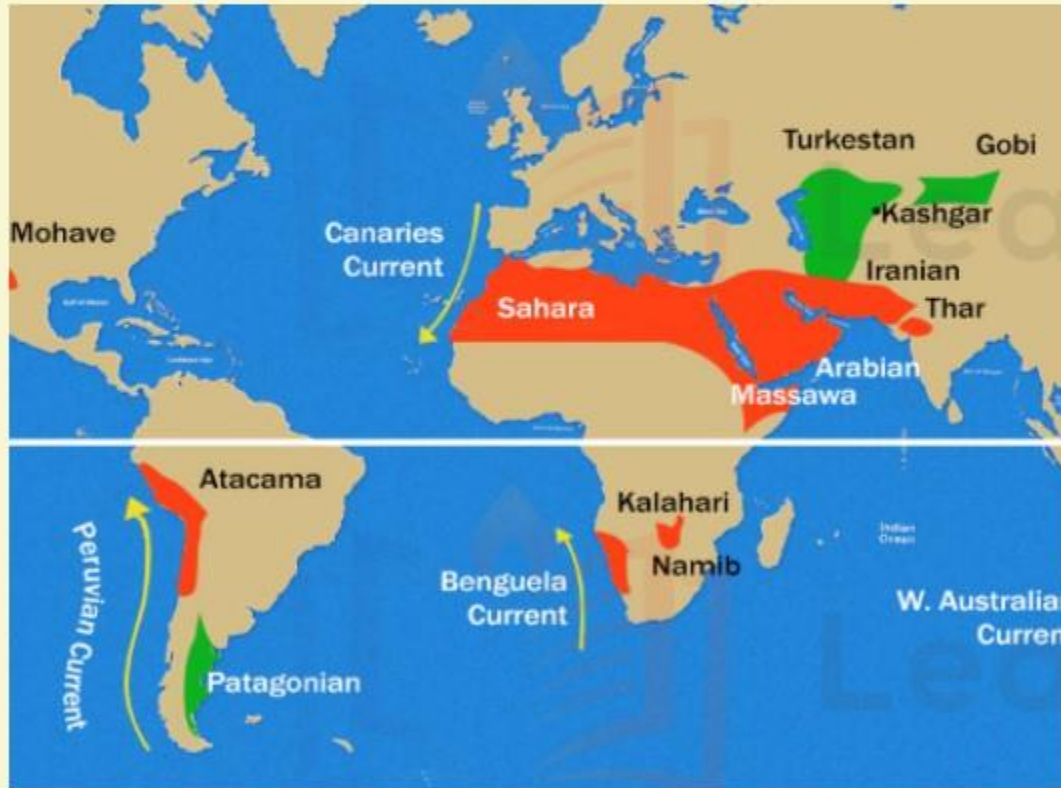
5 In North America, the desert extends from Mexico into USA. It is called by different names or different places like the Mohave Desert, Sonaran Desert, Californian and Mexican Deserts.



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HOT DESERT AND MIDLATITUDE DESERT CLIMATES

DISTRIBUTION

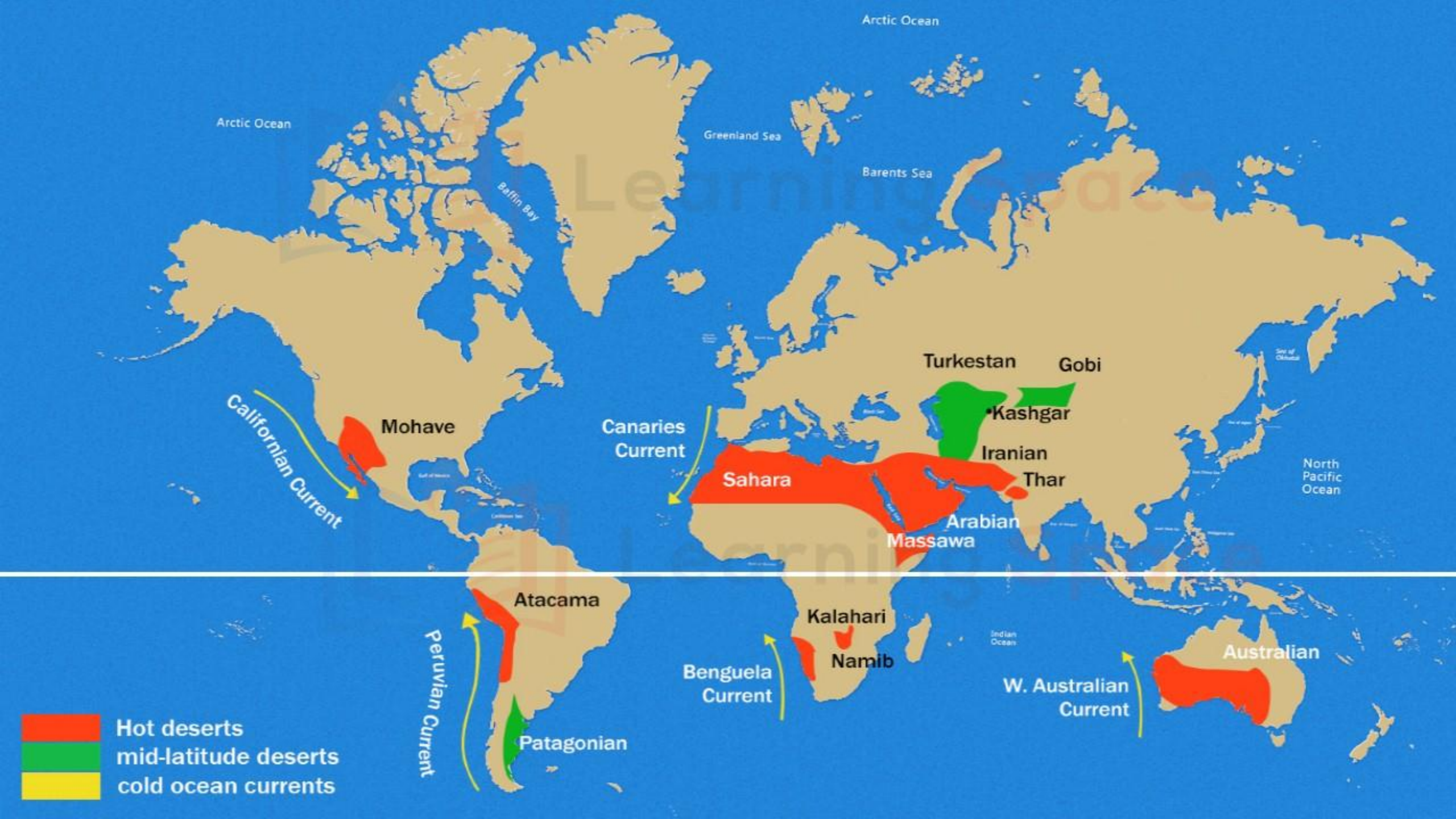


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Amongst the mid-latitude deserts many are found on plateau or at a considerable distance from the sea. These are Gobi Desert, Turkistan Desert and Patagonian Desert.

8

The Patagonian Desert is more due to its rain shadow position on the leeward side of the lofty Andes than to its continentality nature.



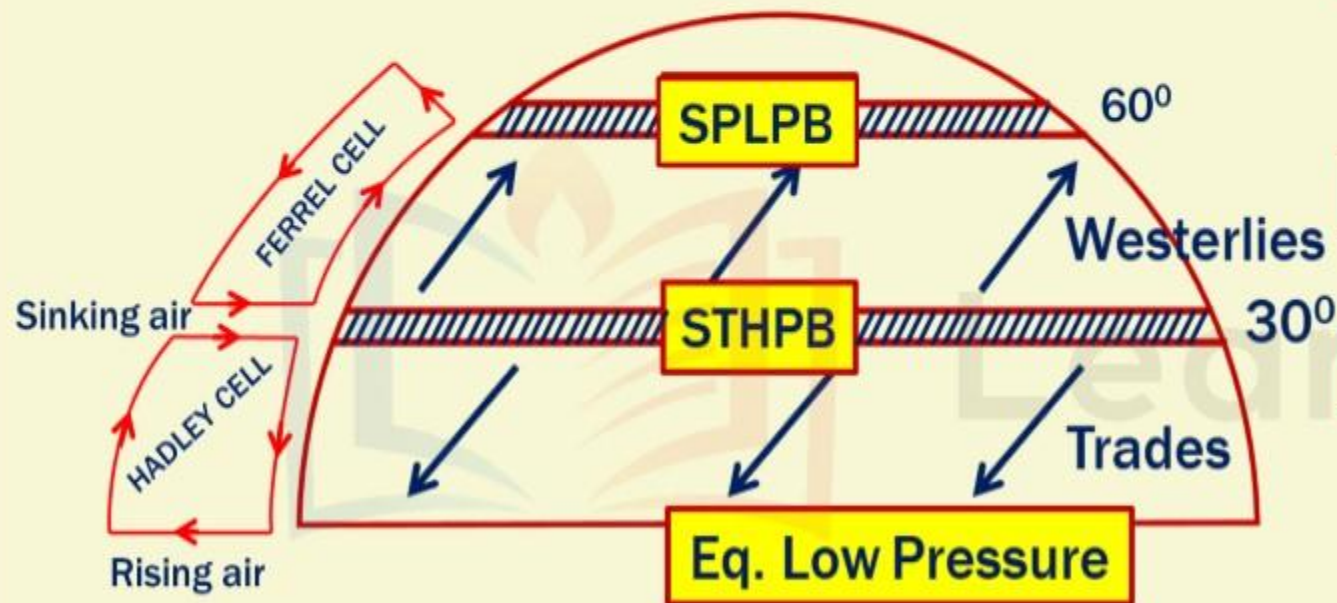
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HOT DESERT AND MIDLATITUDE DESERT CLIMATES

CLIMATE

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RAINFALL



- 1 The rainfall in these deserts are between 2 to 10 inches annually. Very few deserts have more than 10 inches of rainfall.
- 2 The aridity of the deserts is the most outstanding feature of the desert climate.
- 3 The hot deserts lie in the subtropical high pressure belts, where the air is descending, a condition least favourable for the precipitation of any kind to take place.

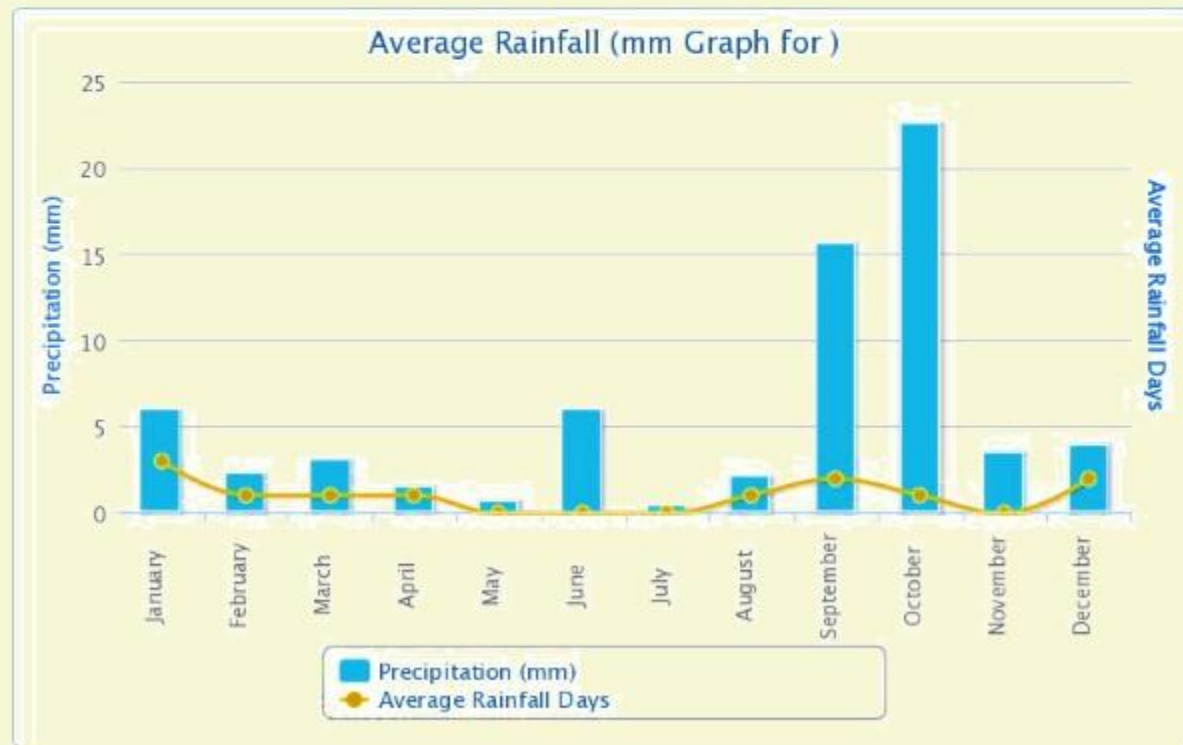
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HOT DESERT AND MIDLATITUDE DESERT CLIMATES

CLIMATE

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RAINFALL



4

The relative humidity of these winds are also considerably lowered because of the hot conditions. Under such conditions every bit of moisture is evaporated and the deserts are thus regions of permanent drought.

5

Precipitation is both scarce and mostly unreliable.

6

On the western coast of the deserts, presence of cold currents give rise to mists and fogs by chilling the oncoming air. This air is later warmed by contact with the hot land and little rain falls.

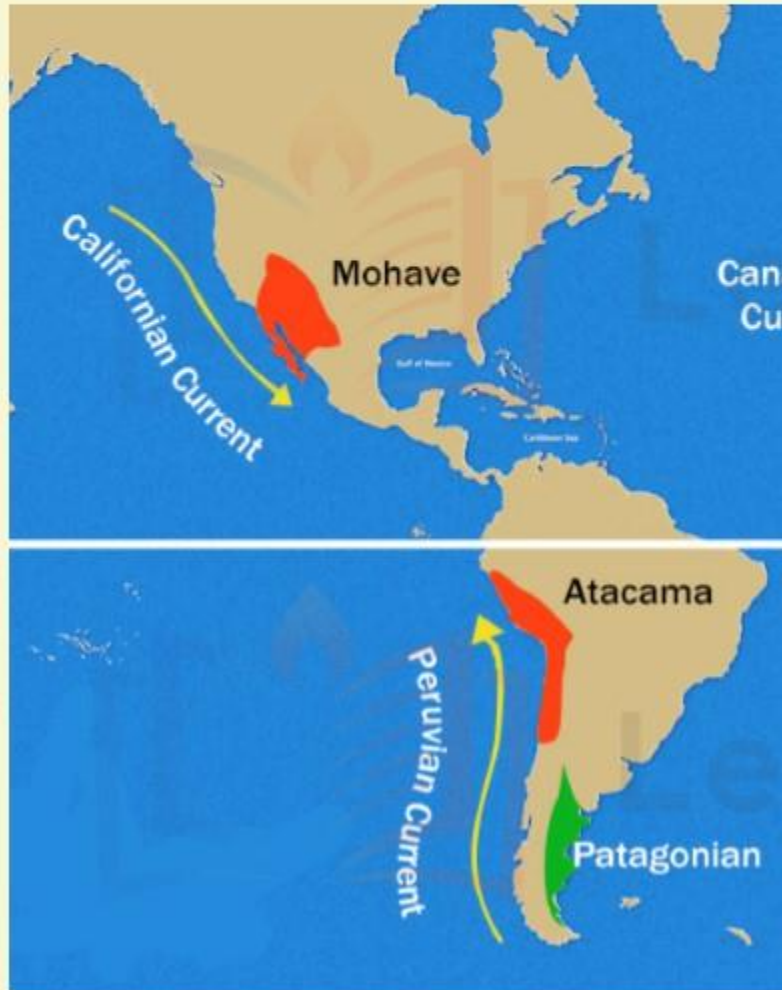
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HOT DESERT AND MIDLATITUDE DESERT CLIMATES

CLIMATE

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RAINFALL



7

The desiccating effect of cold Peruvian current along the Chilean coast is so pronounced that the mean annual rainfall for the Atacama Desert is not more than half an inch.

8

Rain normally occurs as violent thunderstorms of the convectional type.

3

HOT DESERT AND MIDLATITUDE DESERT CLIMATES

CLIMATE

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TEMPERATURE



1

The deserts are the some of the hottest spots on the Earth and have high temperatures throughout the year. There is no cold seasons in the hot deserts and the average summer temperature is 86⁰ F.

2

The reasons for the high temperature is obvious – clear cloudless sky, intense insolation, dry air and rapid rate of evaporation.

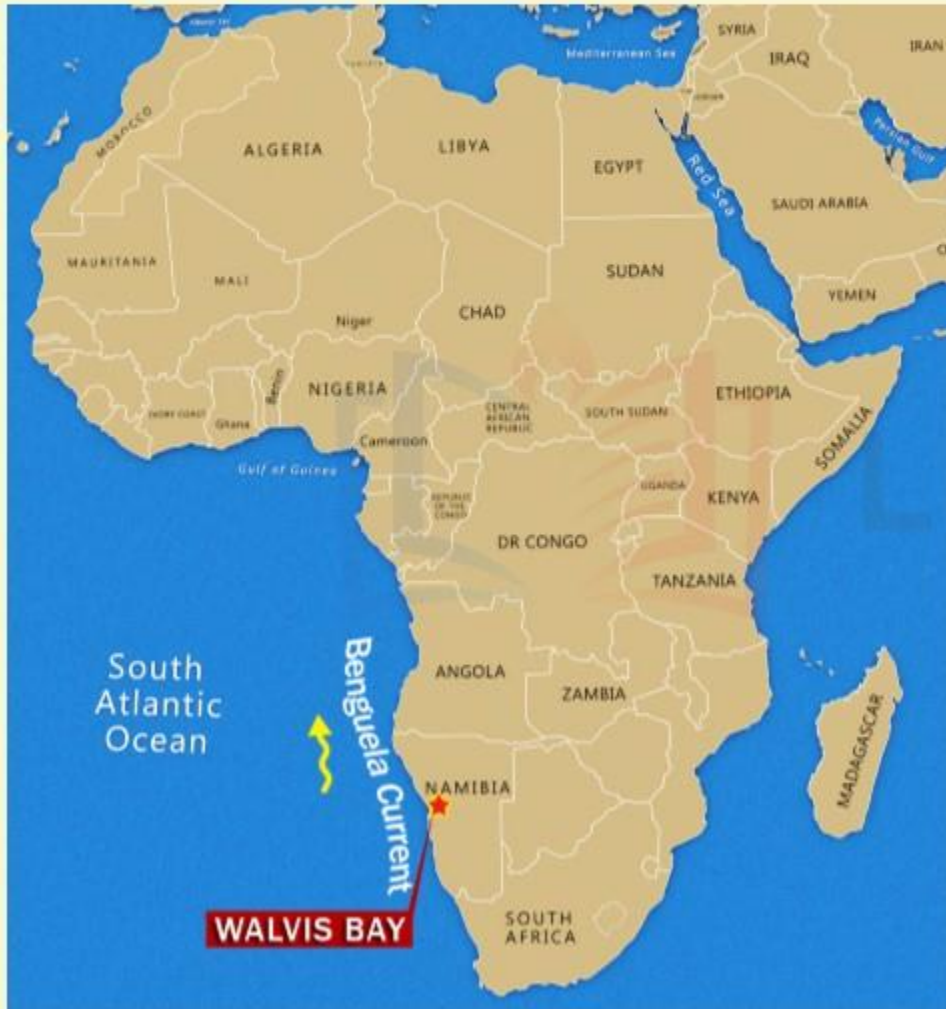
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HOT DESERT AND MIDLATITUDE DESERT CLIMATES

CLIMATE

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TEMPERATURE



3

Coastal deserts by virtue of their maritime influence and cooling effect of the cold currents have much lower temperatures. For eg : **Walvis Bay in South West Africa** has mean annual temperature of only 60 F.

3

HOT DESERT AND MIDLATITUDE DESERT CLIMATES

CLIMATE

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TEMPERATURE

4 The desert interiors however, experience much higher summer temperatures and the winter months are rather cold.

5 For example In Salah in Sahara Desert has a temperature of 99° F in the hottest month and 55° F in the coldest month.

6 Intense insolation in the day causes the temperature rise, but as soon as the sun sets the land losses heat quickly by out radiation and the temperature drops.

7 Thus, it is clear that the diurnal range of temperature in the desert is very great.

8 Frost may occur in the night in the winter. These extreme of the temperatures makes desert living most difficult.

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HOT DESERT AND MIDLATITUDE DESERT CLIMATES

CLIMATIC CONDITIONS IN THE MID LATITUDE DESERTS

1 The climatic conditions of the midlatitudes deserts are many ways similar to those of the hot deserts.

2 Aridity is the key note. The inland deserts lie far away from the sea sheltered by the high mountains all around them.

3 As a result they are cut off from the rain bearing winds. Occasionally depressions may penetrate the Asiatic continental mass and bring light rainfall in winter.

3 Unexpected convectional storms may occur in summer; due to their coldness and elevation snow falls in winter.

3

HOT DESERT AND MIDLATITUDE DESERT CLIMATES

CLIMATIC CONDITIONS IN THE MID LATITUDE DESERTS



4

Continentality accounts for the extremes in temperature. Winters are often severe, freezing lakes and rivers and strong cold winds blow all the time.

5

The greatest inhibiting factors to settlement are the winter cold and the permanent aridity besides the remoteness from the sea.

3

HOT DESERT AND MIDLATITUDE DESERT CLIMATES

DESERT VEGETATION



- 1 All deserts have some form of vegetation, though they may not appear green and fresh all the time.
- 2 The predominant vegetation of both hot and mid latitude desert is **xerophytic and drought-resistant scurb**.
- 3 This includes **cacti, dwarf acacias**. Trees are rare except where there is abundant groundwater to support clusters of date palms.

3

HOT DESERT AND MIDLATITUDE DESERT CLIMATES

DESERT VEGETATION



4 Along the western coastal deserts washed by the cold currents as in the Atacama Desert, the mists and fog are formed by the chilling of the warm air or cold current which nourish a thin cover of vegetation.

6 Intense evaporation increases the salinity of the soil so that the dissolved salts tend to accumulate on the surface forming *hard pans*.

7 Absence of moisture retard the rate of decomposition and desert soils are very deficient in humus.

HOT DESERT AND MIDLATITUDE DESERT CLIMATES

DESERT VEGETATION



- 8 Desert shrubs have **long roots**, which are well spaced out which go deep inside the ground in search of water.
- 9 Plants have **few or no leaves** to reduce the loss of water through transpiration.
- 10 Plants like the cacti have **thick succulent stems** to store up water for long droughts. In short all plants must adapt themselves to survive in such an inhospitable region.

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